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Reverse Engineering Tire Pump Project



Figure 1 Tire Pump

This report examines plug-in tire pump by disassembling into its structural, fluid, and human interface systems including parts such as the pressure gauge, the air compressor, the buttons, structural components, etc. This analysis focuses on gaining understanding of the mechanisms that drive the pumps operation, such as the interaction between the motor and piston or the pressure gauge regulating airflow. The analysis highlights the designs of the pump, but also identifies improvements that could be made to improve the product. This study emphasized the importance of the integration of fluids, structures, and human interface in products like these and identifies ways for new design improvements.

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Introduction

Tire inflators are essential tools that ensure proper tire pressure, which is critical for vehicle safety, fuel efficiency, and extending tire life. Whether at home or in professional settings, these devices provide a quick and reliable way to inflate tires, making them indispensable for drivers and mechanics alike. Through reverse engineering, we can take a closer look at the design and operation of a common device like the Hyper Tough 12V Standard Tire Inflator. This process not only helps us understand how it works but also highlights potential improvements to enhance its functionality and usability.

This project focuses on three primary subsystems: fluids, structural and mechanical components, and user interface (UI). The fluid subsystem manages air compression and pressure regulation, the structural and mechanical subsystem supports the device and ensures efficient power transmission, and the UI subsystem facilitates user interaction through controls and displays. By analyzing these subsystems, we aim to better understand how the inflator functions and identify opportunities to enhance its performance and user experience.

Methodology and Procedure for Deconstruction

These steps go through the deconstruction and disassembly of the tire pump and its components.

1. Using a phillips-head screwdriver, unscrew the 4 screws on the back plate.
2. Then, separate the front plate and back plate.
3. Next, unscrew the interface panel from the front plate.
4. Then, disconnect the spring from the button from the interface plate.
5. Then, cut the wires that lead from the motor to the spring behind the UI system and the wires that lead from the motor to the circuit board for the light.
6. Cut the pipe that leads to the pressure gauge.
7. Cut the air hose where it meets the piston.
8. Remove the fan from the motor axle using brute force.
9. Remove the splitter for the two air pipes and remove the cap from the motor.
10. Remove the 2 screws and separate the gear and piston from the motor.
11. Remove the piston from the plastic gear by prying the bearing off the connection.

Functional Decomposition Chart

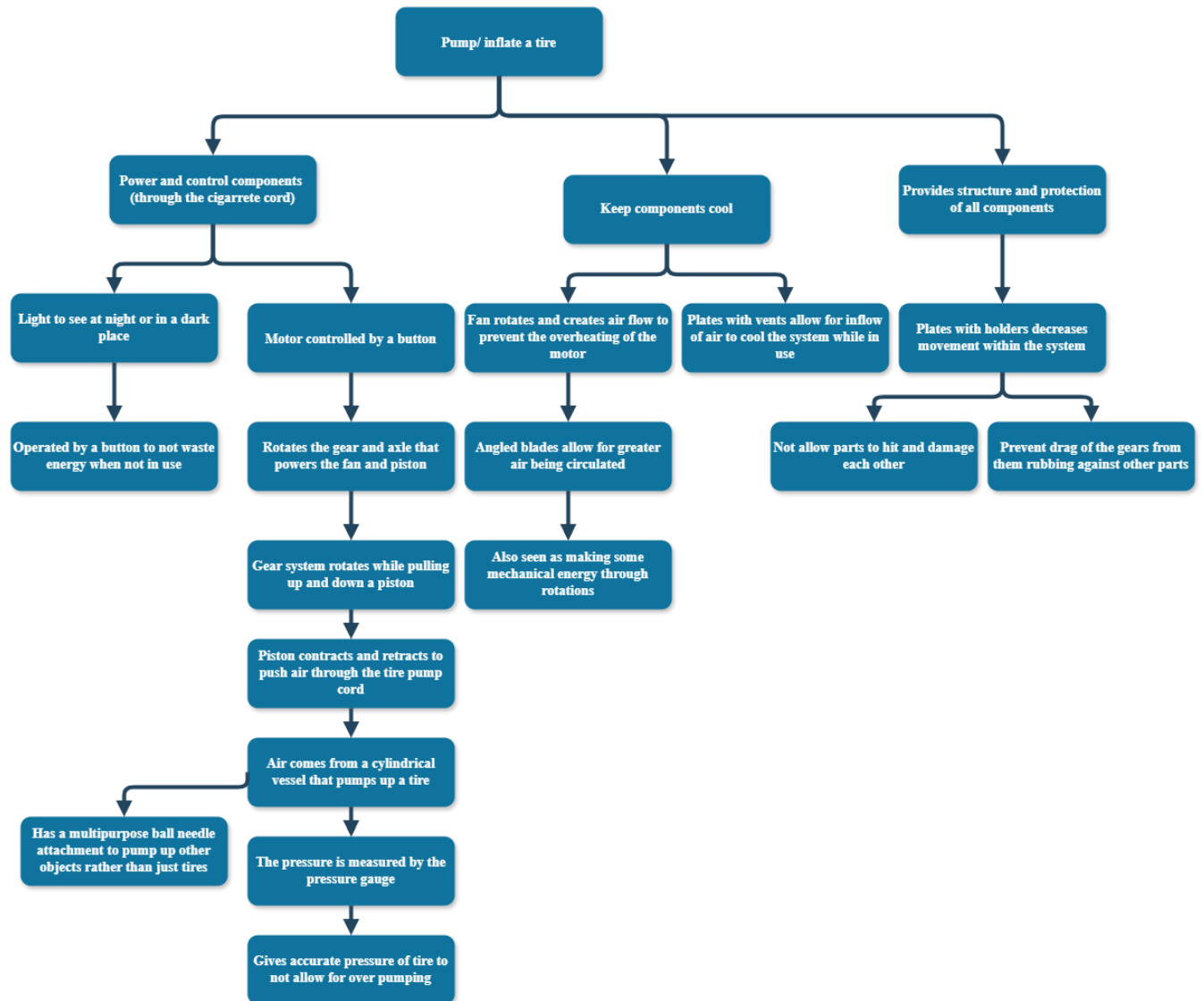


Figure 2 Functional Decomposition Chart

The main overall function of the tire pump is to inflate a tire. This can be broken down into three subfunctions: power and control components, keep components cool, and provide structure and

protection of all components. The components are kept cool through the use of vents on the sides of the plates and the circulation of air by the fan. There are cradles and holders built into the plates that allow the components to have minimum movement throughout use. The cigarette cord provides power for the light and motor, which is the main driving force. It creates motion in the axle that is connected to the metal gear and fan. The metal gear then rotates the plastic gear and pulls the piston along to compress air through the cylindrical vessel to supply air to the tire. On the other side of the motor, the fan is rotated at the same rate and provides cooling for the entire system when in use. There are also vents in the outside plates that allow for the flow of air. A built in flashlight allows the user to see better at night or in a dark place. Most of the components are made of a hard plastic except for the motor, axle, and smaller gear which are made out of metal.

Gantt Chart

PROJECT: Tire Pump Reverse Engineering ME 1212 Table 6

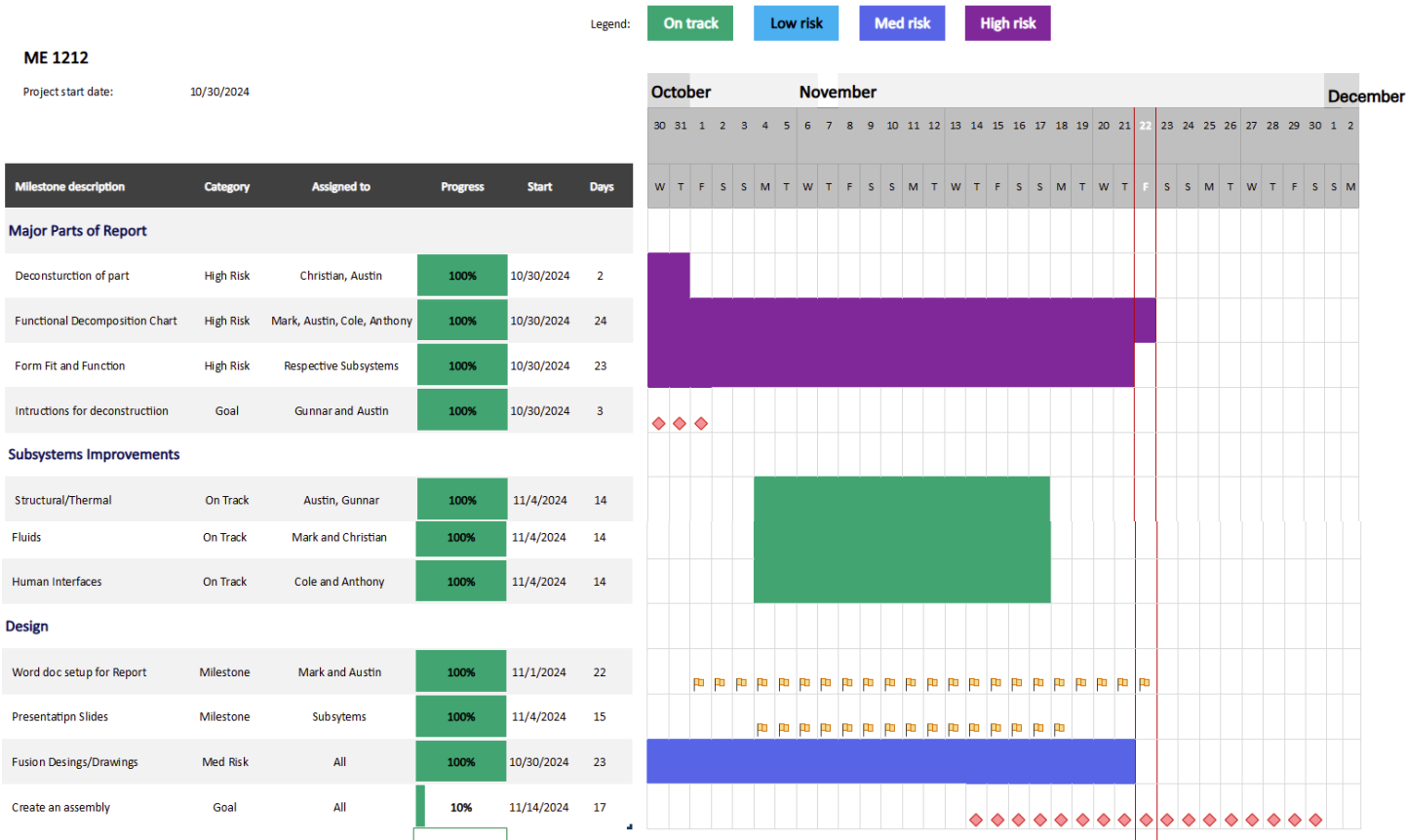


Figure 3 Gantt Chart

Form, Fit, and Function

Structural/ Thermal

Fan



Figure 4 Tire Pump Fan

Weight 1.64g

Form

The fan has a cylindrical center with blades that are slanted that rotate around an axle.

Fit

The fan has an open hole in the middle cylinder to connect to the motor by an axle which will allow it to spin.

Function

The fan rotates as the motor spins the axle causing the fan to create air flow to cool the whole system down to prevent the motor and inner components from overheating. The blades being angled allows for greater circulation of air.

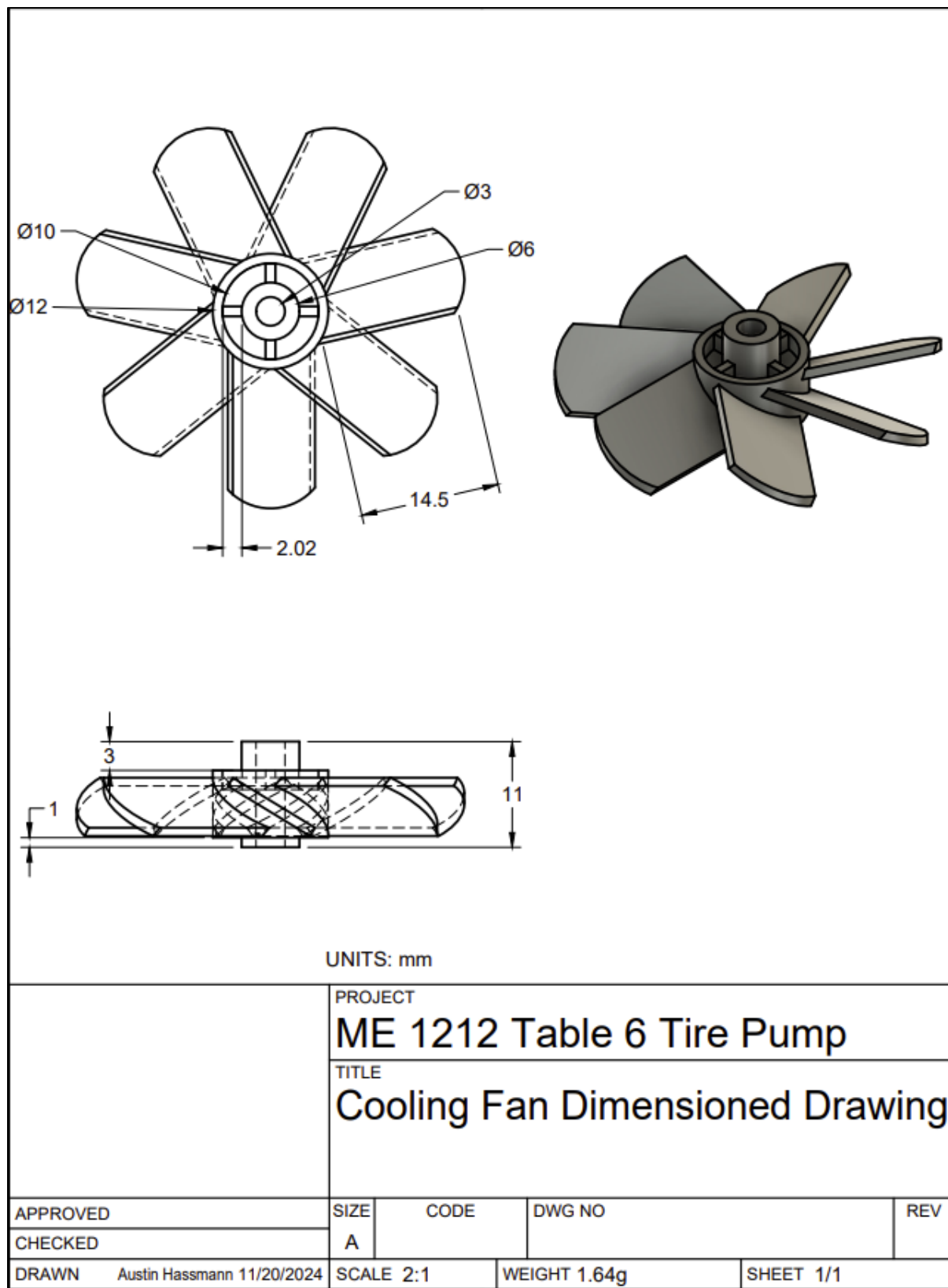


Figure 5 Fan Dimensioned Drawing

Back Plate



[Figure 6](#) Back Plate Inside View



[Figure 7](#) Back Plate Isometric View

Weight 121.43g

Form

The back plate is the outer shell in the form of a rounded off square with indentions and brackets on the inside for parts.

Fit

The back plate is designed to hold all the components of the system in an organized way. It also screws into the front panel to secure the whole system. The circular holder contains the motor and cylinder which allows for the gear and fan to be supported above the shells.

Function

The back plate serves as the backbone for the system and can be a protective shell for the parts and system as a whole. It secures the inner parts and decreases the movement they could have and any energy wasted from the inner parts being loose.

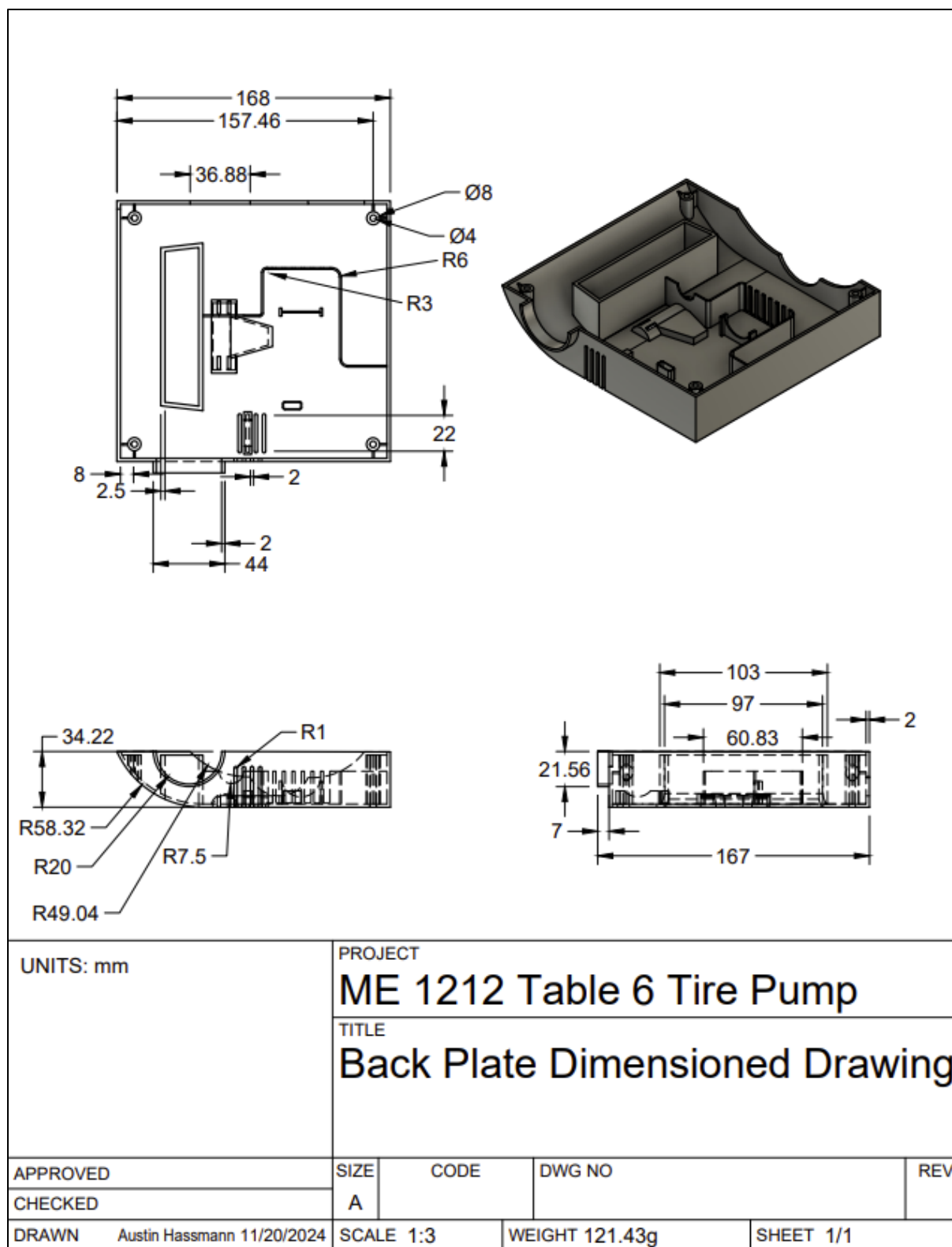


Figure 8 Back Plate Dimensioned Drawing

Vents



Figure-9 Back Plate Vent

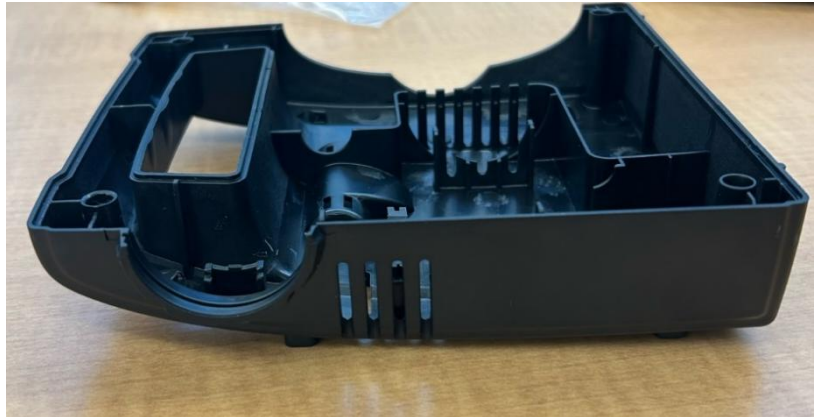


Figure 10 Vent Side View

Form

The vents are slits in the sides of the plates in the form of rounded off rectangles.

Fit

The vents are cutouts from the original plates like rectangular holes.

Function

The vents allow for greater airflow throughout the whole system by allowing air to come into the system to cool off the motor and components.

Motor



Figure 11 Motor Isometric View

Weight 117.8g

Form

The motor is a cylindrical component made from metal with an axle going through it.

Fit

The motor is attached to the gear and piston by screws on one of the sides. The other side has an axle that connects to the fan.

Function

The motor rotates the axle that spins the fan to cool the entire system. It also rotates the gear which pushes and pulls the piston to create a pump of air through the hose and therefore pumps up the tire.

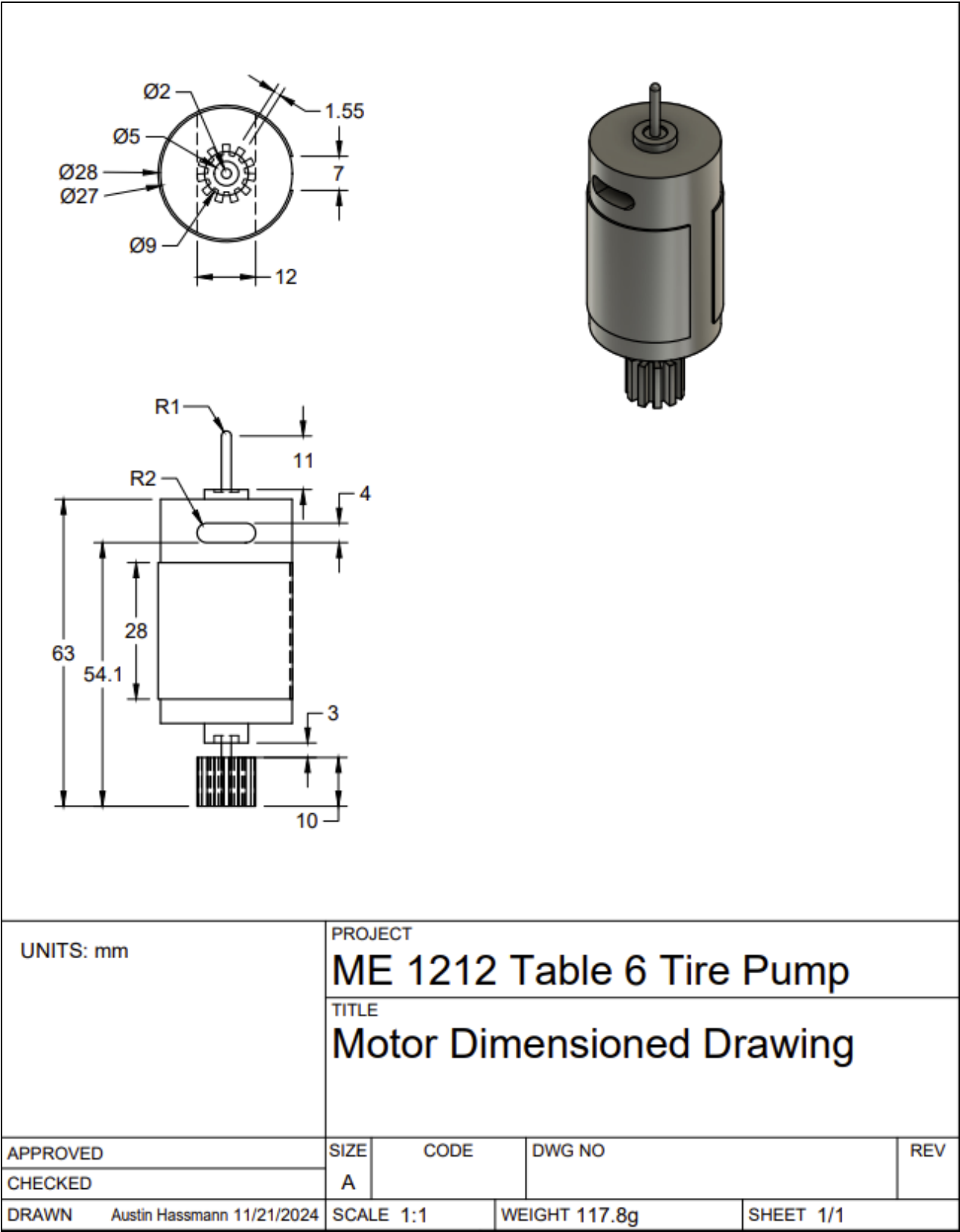


Figure 12 Motor Dimensioned Drawing

Front plate



Figure 13 Front Plate Button Interface



Figure 14 Front Plate Top View

Weight

Form

The front plate is modeled as a base for which all the parts can fit neatly and snug. It has a vent for the fan, a cradle for the motor and the piston, guiding cradles for the different pipes, and a cradle for the light at the top right of the pump. There is a handle molded into the shell, as well as a large opening in the front that is used to hold the outer casing for the buttons used to activate the tire pumping process.

Fit

It has 4 threaded holes at each corner for the back plate to screw into it. The large opening on the front face leaves room for another, smaller piece to be fitted into it that holds the buttons and pressure gauge at the front.

Function

This part has the purpose of keeping the inner working parts in place and protecting them from damage.

Figure 15 Front Plate Dimensioned Drawing

Metal Gear

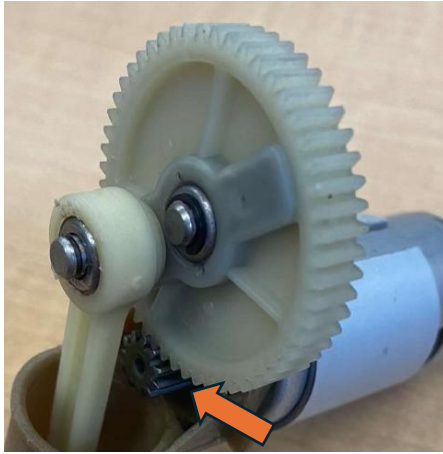


Figure 16 Metal Gear on Motor Axle



Figure 17 Metal Gear Isometric View

Form

The metal gear is 11 toothed, about a third of an inch long and attached to one end of the motor.

Fit

The metal gear is connected to the axle of the motor that fits to the plastic gear.

Function

The metal gear rotates to move the plastic gear so that the piston moves up and down to compress the air through the air hose

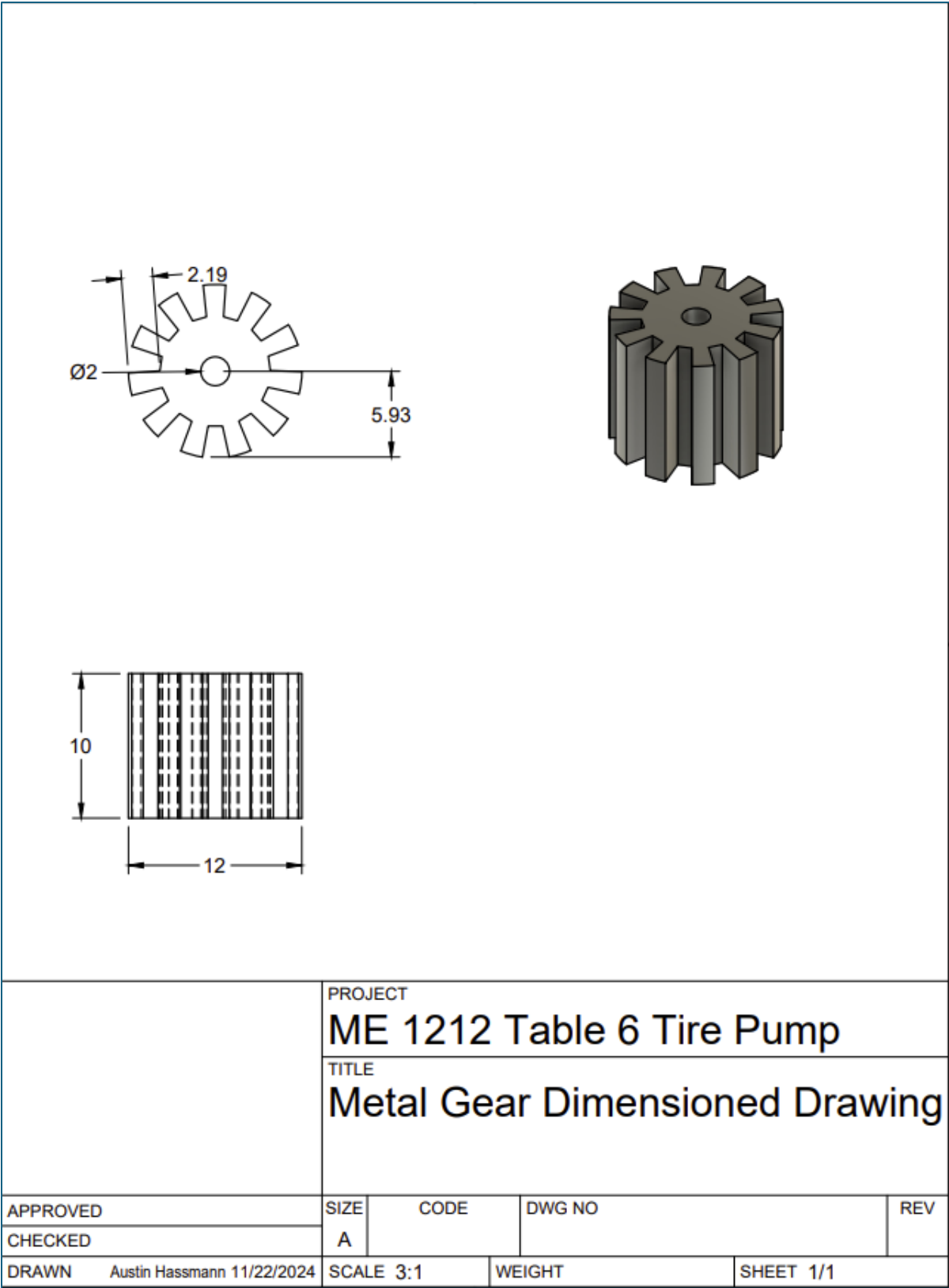


Figure 18 Metal Gear Dimensioned Drawing

Interface Plate



Figure 20 Interface Plate Top View



Figure 21 Interface Plate Isometric View

Weight

Form

It has a conic curve shape and a hole in the middle for the pressure gauge.

Fit

The interface plate is a part that snaps into the front plate to house the interface buttons. There is a spring behind the power button, and the light button is held by a thin piece of plastic that is a part of the plate itself.

Function

Those used to turn on the pump and to toggle the light on and off.

Human Interfaces

Button



Figure 22 Button Isometric View

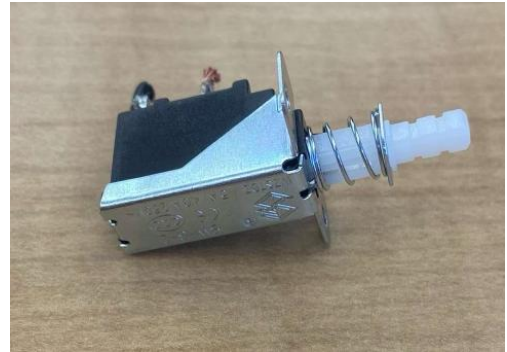


Figure 23 Button Front View

Weight 5.74 g

Form

The button is rectangular shaped. It has a protruding white square piece at the front with a metal spring wrapped around it and is held in place by a metal plate fitted around the white square.

There is a metal cover that covers the lower half of the body and allows it to be fastened into place. Finally, on the top of the body lies two nodes used for wiring.

Fit

The button hides behind a small, circular face plate fitted into the front plate of the tire pump. As the user presses upon the plate fitted in front of the button, the button is pushed inwards by the face plate and locked into place. The button switch is connected by 3 wires connecting it to the tire pump system. Two wires are conjoined to the back node and one wire is connected to the button's front node.

Function

The button allows for the user to activate and deactivate the entire tire pump system. As the user pushes against the plate in front of the button, the button is pushed inwards and locked into place, which activates the tire pump system. The tire pump system will run as long as this button is locked in its compressed state. To deactivate the system, the button is pressed once again, which unlocks the button, releasing it outwards into an extended state.

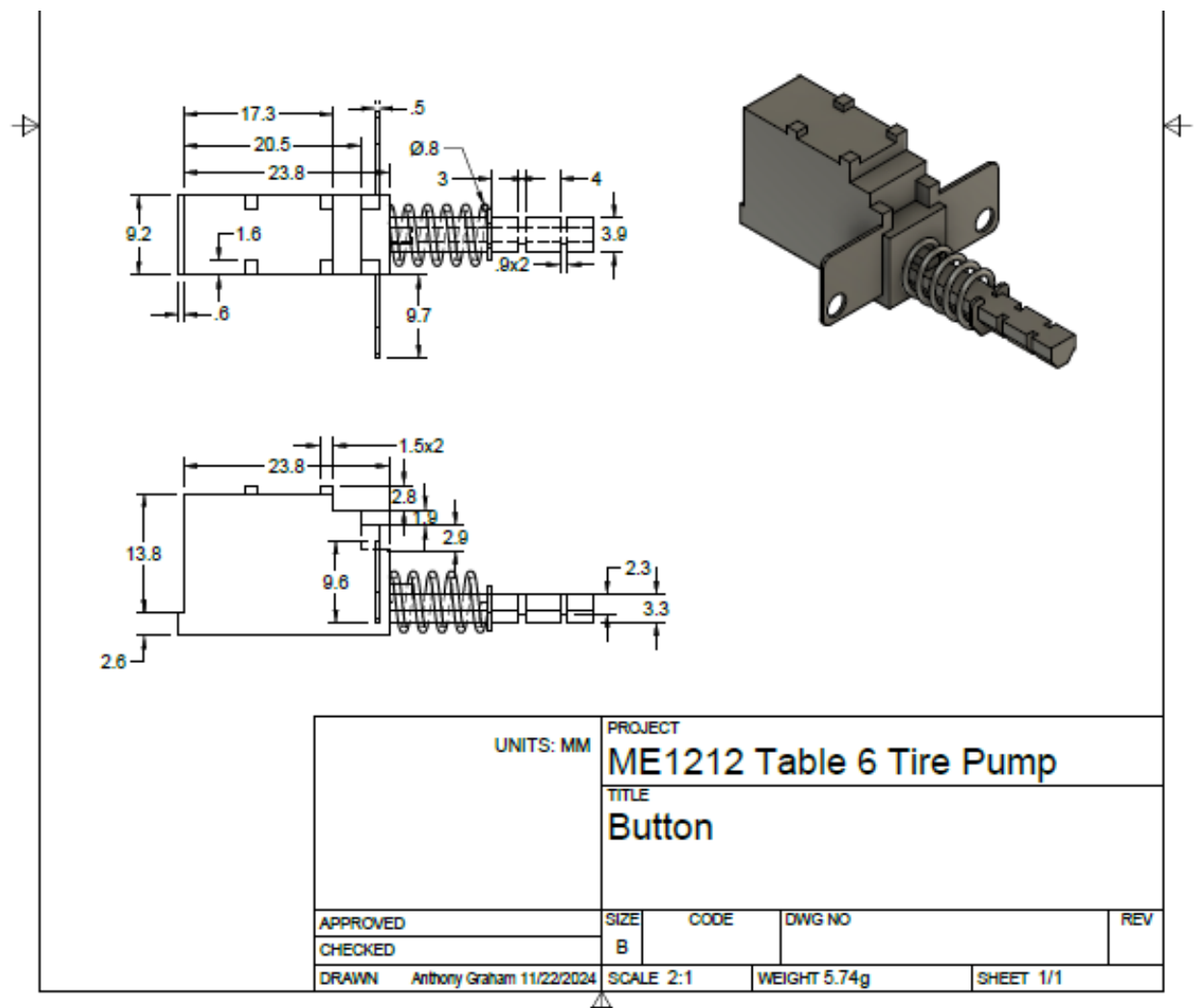


Figure 24 Button Dimensioned Drawing

Car Plug in



Figure 25 Car Plug In Top View



Figure 26 Car Plug In Isometric View

Weight 73.42g

Form

The car plug-in component of the electric tire pump device has a black, modified cylindrical shape. Being made almost entirely out of plastic material, it is very portable along with the rest of the car pump device. One end of the plug-in piece has a small metal piece that is used as a conductor and the other end holds the cord for the plug-in to connect to the tire pump.

Fit

The car plug-in component serves as an extension of the tire pump device. It connects to the main device through a cord measuring roughly twelve inches long (1 foot). The cord extends out from the left side of the main device. There is no mount on the main tire pump device for the car plug-in end piece as it is normally plugged into an automobile to provide electricity to the device.

Function

The plug-in device acts as a very important part of the entire tire pump system as it provides electricity to the pump. While it is plugged into a port inside an automobile, the plug-in provides a constant power supply for the pump to be used to pump any deflated tire. Without this essential part of the device, there would be no source of power or source of power collection for the pump to operate at all.

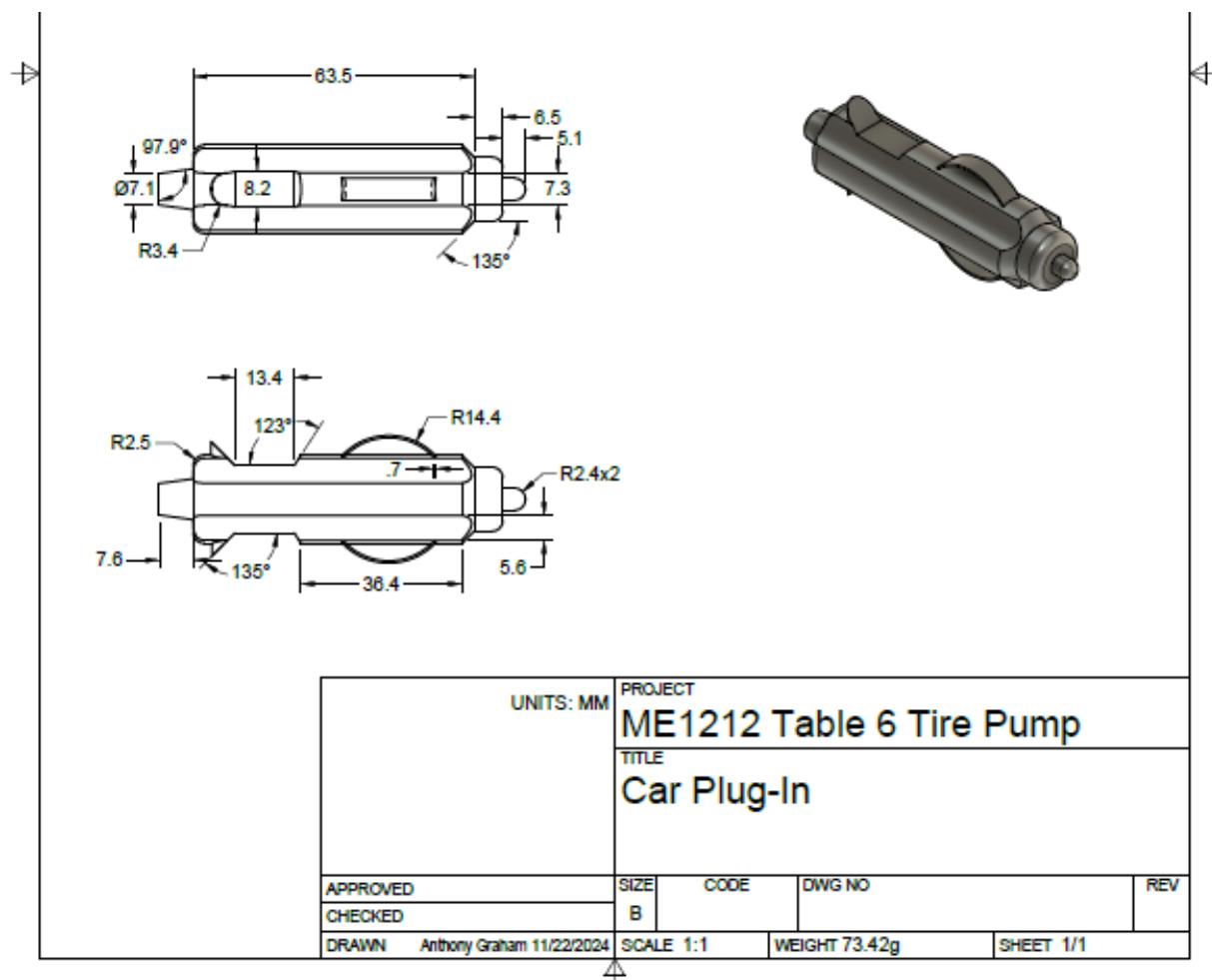


Figure 27 Car Plug-In Dimensioned Drawing

Valve



Figure 28 Valve Isometric View

Weight 1.24g

Form

The screw-on valve piece has a very simple cone shape, is completely black, made of plastic, and extremely lightweight. At the wide end of the cone, there is a cylindrical threaded portion extending out that is to be used for the valve to be screwed onto the main tire pump device.

Fit

The threaded portion of the valve aids in fitting onto the main device. It is screwed onto the device as an extension and assists with air flow from the device.

Function

The valve functions as an extension piece that can either be added or removed to the overall functionality of the electric tire pump. When removed, the option for using the ball needle as an air outlet is made available. When screwed into the extending hose coming from the main device, the valve can be used as an alternative outlet method for airflow.

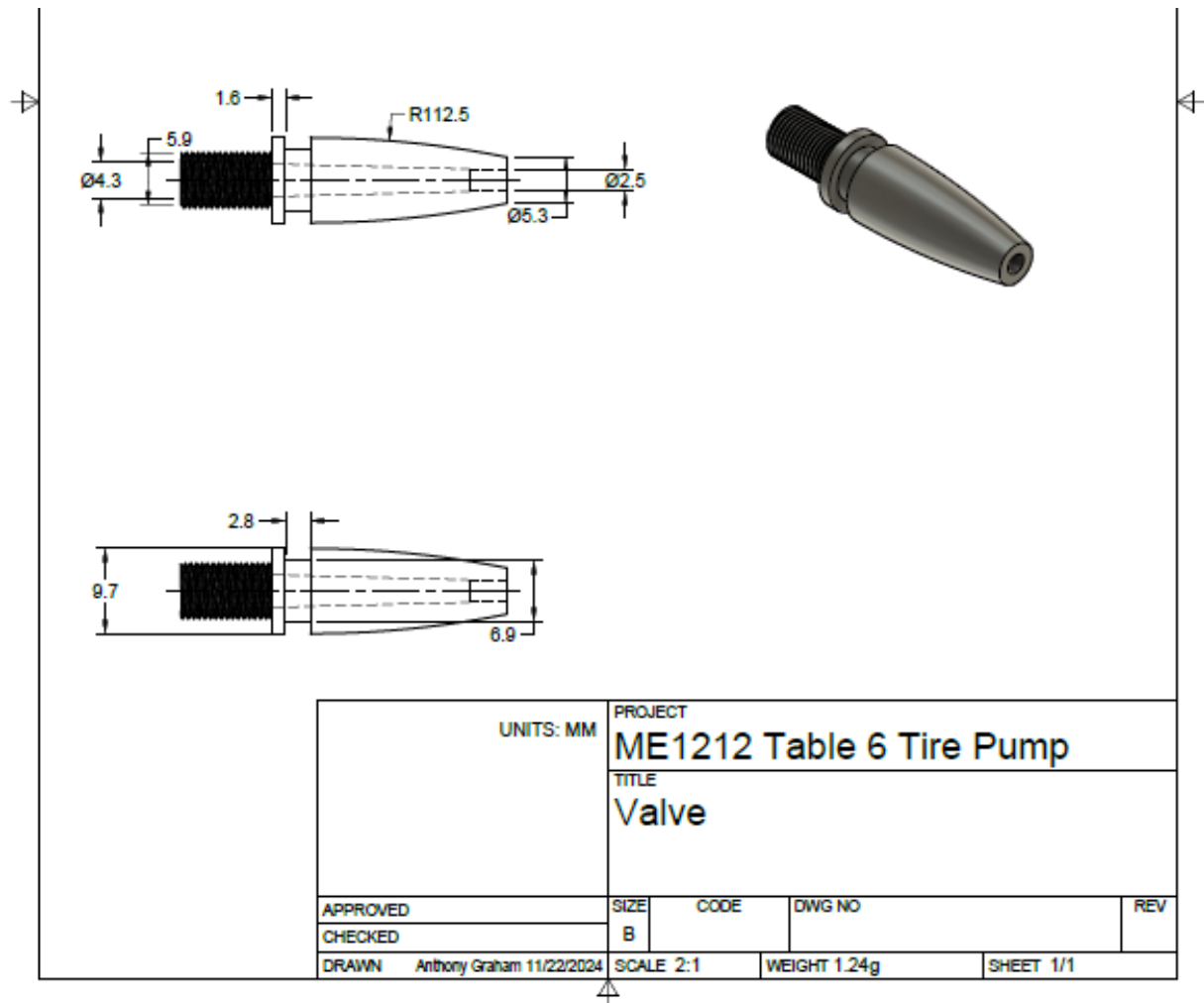


Figure 29 Valve Dimensioned Drawing

Handle



Figure 30 Handle on whole system

Form

The handle on top of the pump is made out of plastic and cut out of the two plates.

Fit

The handle is cut out of the pump's shell and creates a place for a human hand to fit inside when the shells are enclosed together.

Function

The handle or cut-out on top of the pump is used to make it easier for the user to pick up and move around.

Flashlight



Figure 31 Flashlight within the plates

Form

The light is made out of a clear plastic material and is housed in a circle cut out on the side of the pump.

Fit

The light fits in the side of the pump in a plastic casing that houses the three light bulbs in place.

Function

The flashlight helps light everything up so it makes it easier for the user to see what they are looking at especially at night time. The button on the front plate controls this feature and makes sure there is no wasted electricity when it is not needed.

Light Casing



Figure 32 Light Casing Top View

Weight 3.70g

Form

The light casing is made out of plastic and two clips on the side of it to snap in between the front and back plate of the pump so it fits snugly.

Fit

The light casing snaps between the pump's front plate and back plate and holds the three light bulbs in place.

Function

The main function of the light casing is to hold the three tiny light bulbs and the circuit board in place.

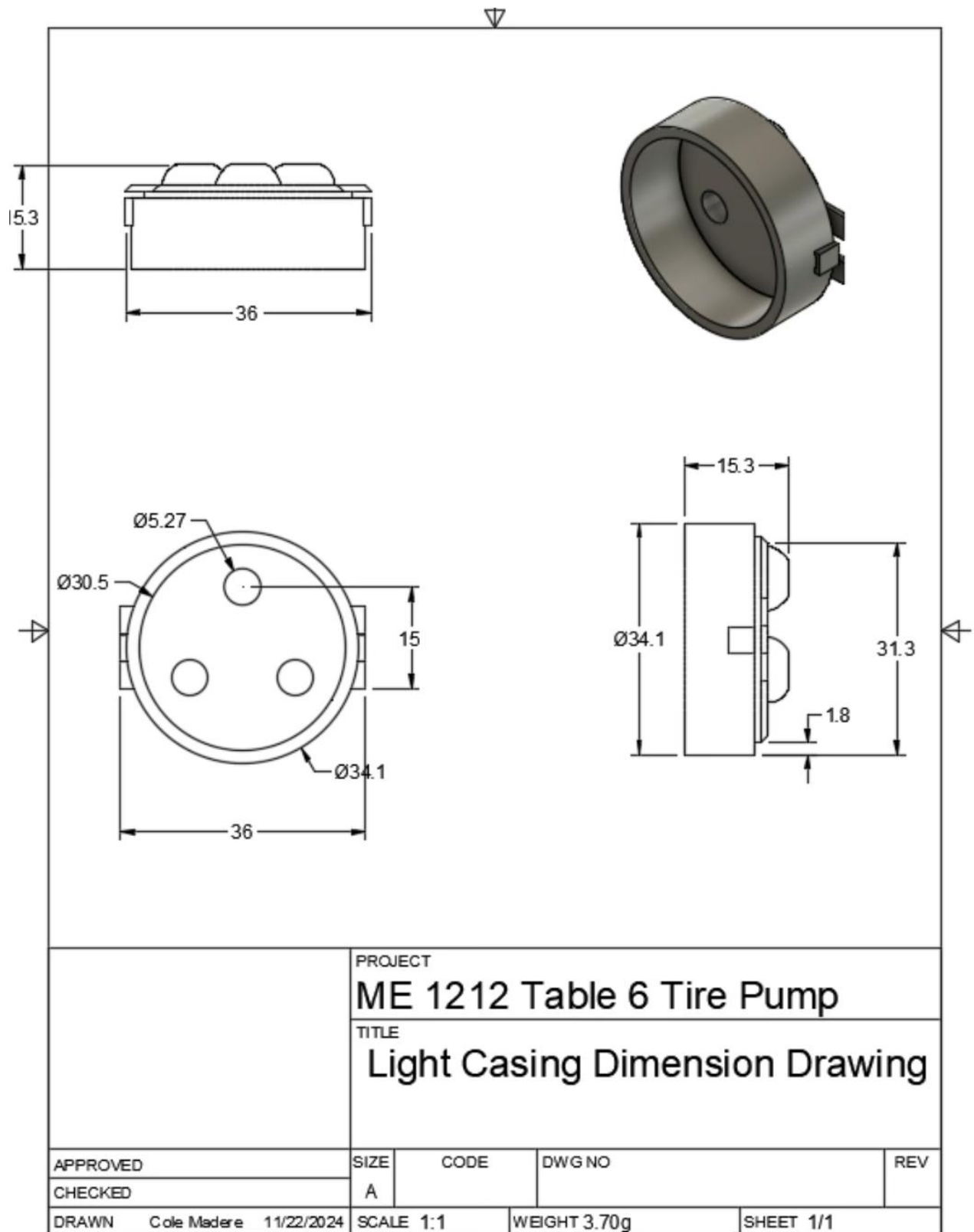


Figure 33 Light Casing Dimensioned Drawing

Flashlight Lens



Figure 34 Flashlight Isometric View

Weight 2.57g

Form

The flashlight lens is entirely made of clear plastic.

Fit

The lens has two notches on the top, bottom, and sides of the lens that fit into notches on the pump shell and the light casing.

Function

The flashlight lens's main function is to protect the light pumps, so when the pump is thrown around or dropped it adds a layer of protection, so there is less of a chance of the bulbs breaking.

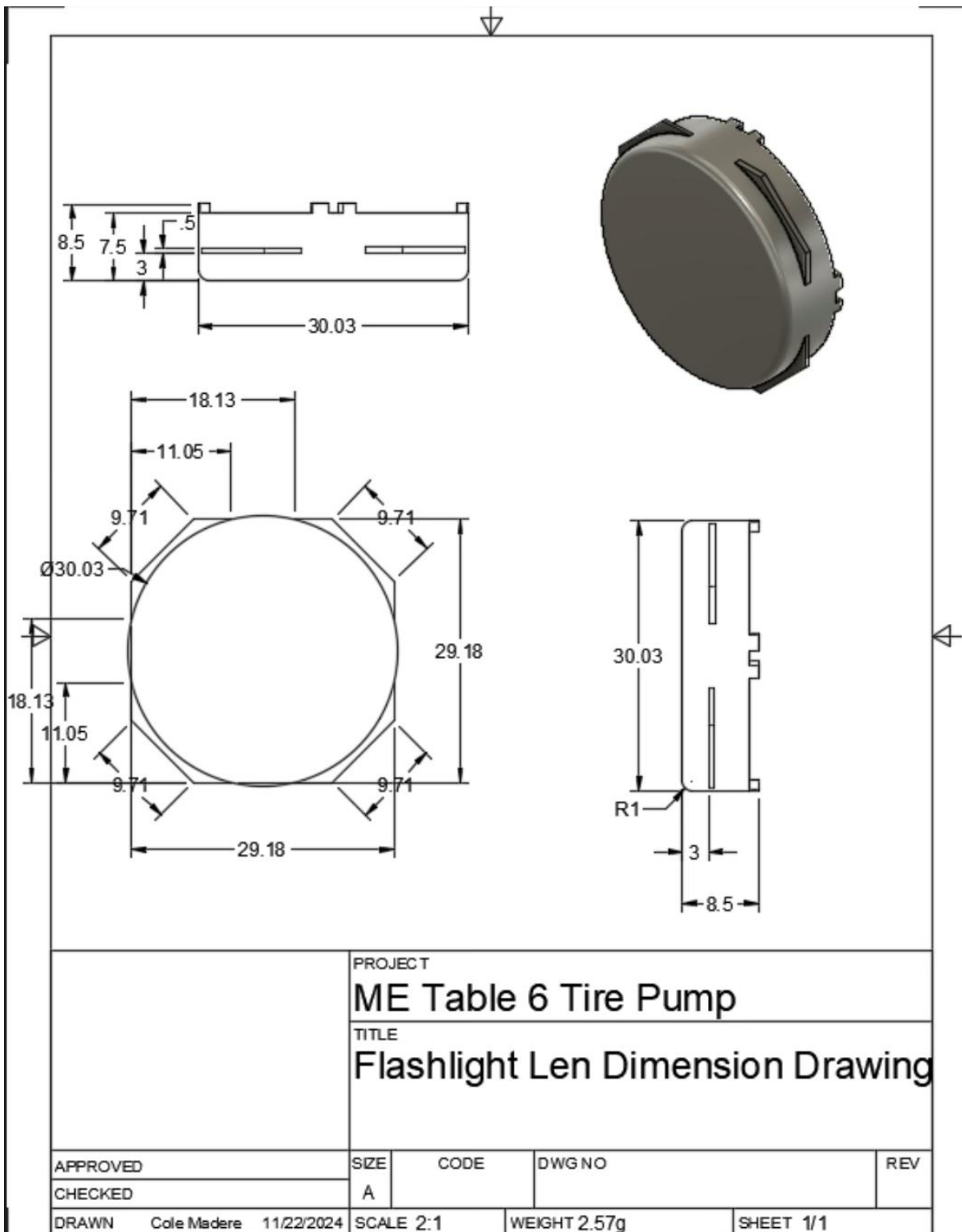


Figure 35 Flashlight Lens Dimensioned Drawing

Ball Needle



Figure 36 Ball Needle Top View

Weight 1.88g

Form

The ball pump attachment is entirely made out of metal with rounded edges that make it safe and easy to use.

Fit

The ball pump needle is an attachment that screws into the end of the hose and is used to inflate a ball quickly

Function

The ball pump needle's main function is to inflate and deflate a ball and make the tire pump more versatile.

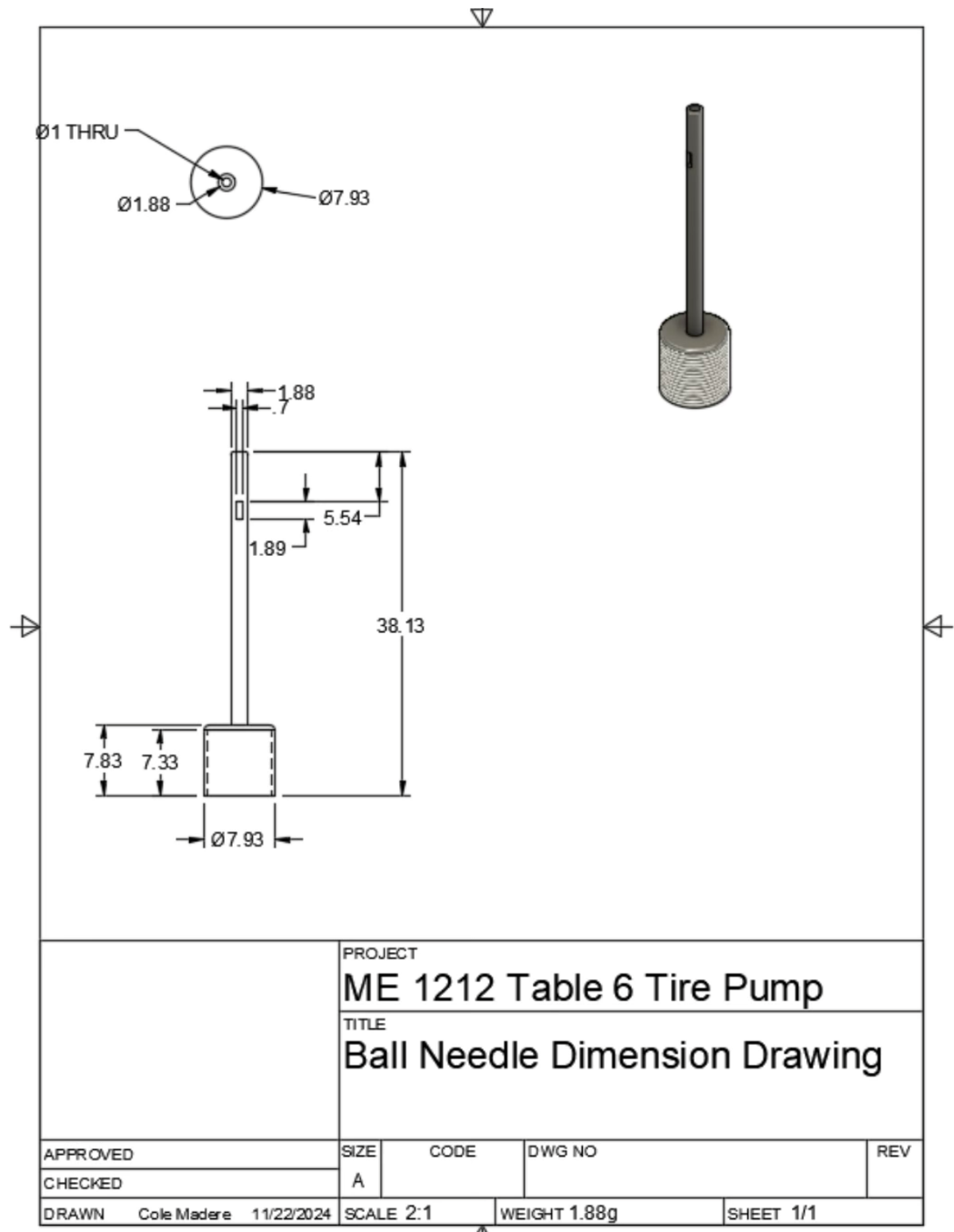


Figure 37 Ball Needle Dimensioned Drawing

Fluids

Piston



Figure 38 Piston Front View

Weight 6.20g

Form

The piston is made from hard plastic, with a slightly concaved base and smooth base to minimize friction. It also has a vertical rectangle with a circular top.

Fit

The circular top of the piston is made to connect to the gear and the base is made to fit in the tube/cylinder where the piston compresses air.

Function

The piston compresses air with the help of the gear system. That compressed air then goes through an outlet at a certain pressure to pump up a tire or object.

Figure 39 Piston Dimensioned Drawing

Hose



Figure 40 Hose

Weight 50.59g

Form

The hose is made of durable rubber with a braided exterior for added strength and grip. It is flexible, and just the right size at 30 inches for convenient use and optimal air pressure.

Fit

The hose is designed to be securely connected to the pump itself and to whatever valve an object has. Due to the hose's flexibility as well, it can fit inside of the shell of the pump for storage.

Function

The hose facilitates efficient air transfer from the pump to the tire, it withstands high-pressure airflow, and remains durable under regular use.

Figure 41 Hose Dimensioned Drawing

Hose Connector



Figure 42 Hose Connector Isometric View

Weight 4.59g

Form

The hose connector is made from hard and durable plastic to ensure its function after heavy use.

It has two connection points: one pointing downward and another on the side. The one of the side features a triangular tip to securely attach the gauge tube

Fit

The connector fits between the main hose and gauge tube, ensuring a tight seal to prevent air leaks and secure attachments to both parts

Function

The hose connector facilitates the transfer of compressed air from the pump to the hose and gauge, allowing the gauge to accurately measure the air pressure, and allowing the hose to receive enough compressed air

Figure 43 Hose Connector Dimensioned Drawing

Pressure Gauge



Figure 42 Pressure Gauge

Weight

Form

The gauge is a circular object with two rounded-rectangular objects with holes on each side directly across from each other. Additionally on the back is a hexagon shaped object with an opening for a hose.

Fit

The pressure gauge is fitted with the air compressor within a hole (approximately the size of the pressure gauge). It is also attached to two different hoses used to direct airflow with an unknown adhesive and a metal wire that is thoroughly wrapped around the connection

Function

The purpose of this item is to provide information of the PSI of what the air is being inserted into. The pressure gauge is attached to a hose that runs to another hose that is used to push air into the desired item. With the air hose connected to the desired item, it connects to a curved tube that is moved when exposed to pressure causing the dial to rotate that indicates PSI.

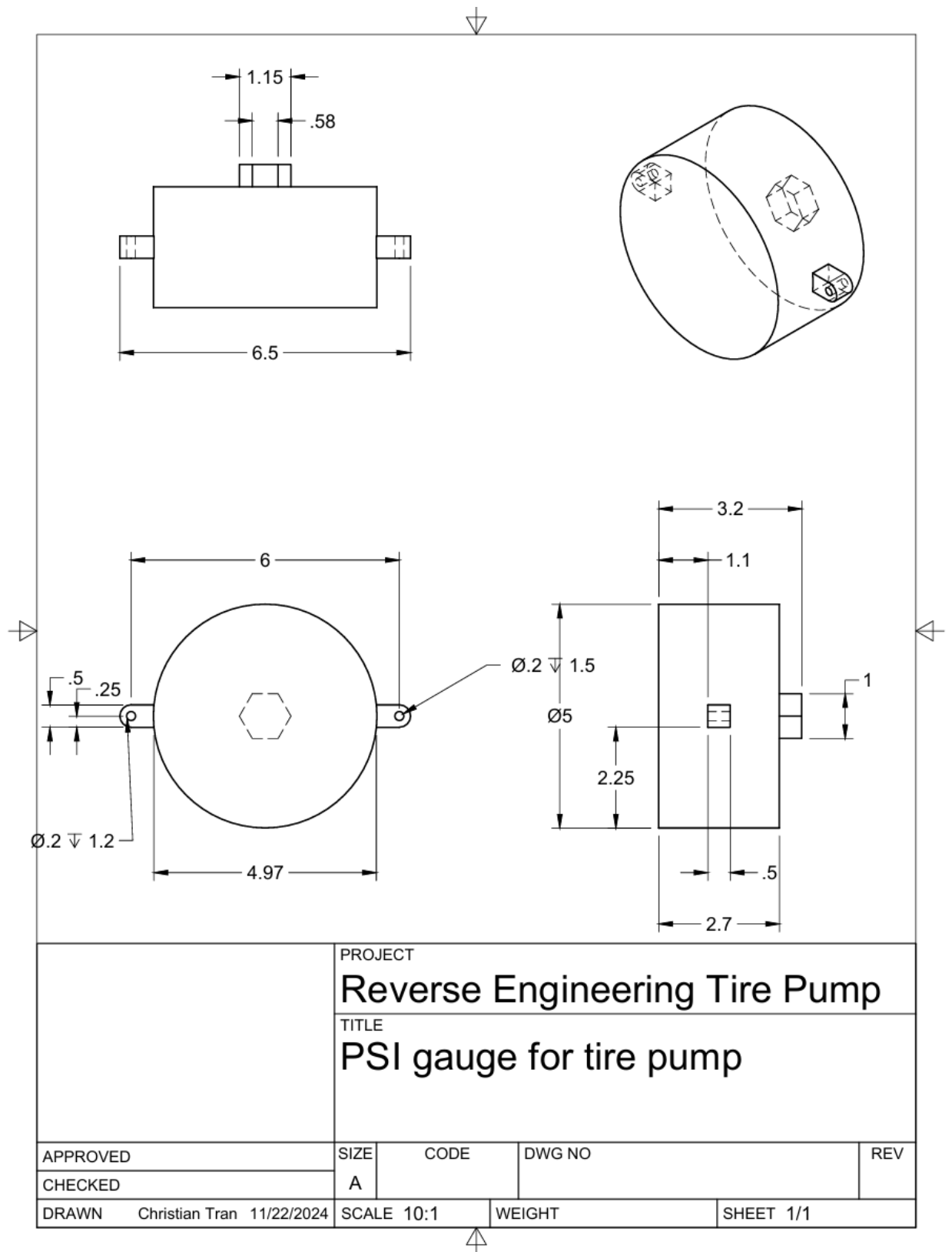


Figure 43 Pressure Gauge Dimensioned Drawing

Plastic Gear



Figure 34

Weight

Form

The gear consists of approximately 54 teeth in the shape of triangles. Along the edge of the gear is a metal cylinder/axle.

Fit

The gear is fitted with an axle running through a hole within the gear with a small pin like piece used to keep the gear on the axle. The gear also has a small axle near the edge for the piston to attach to.

Function

The function of this gear is to be a main hub for movement of a piston. The gear contains an axle located close to the edge allowing for an up and down like movement caused by the rotation of the gear.

Air component



Figure 37



Figure 36

Weight

Form

The air component consists of a cylinder that has two different additional cylinders that are progressively smaller towards the bottom. There is a beam running on top of the main cylinder with a hole at the top which is the same size as the hole in the gear.

Fit

The air component is attached to the motor via screws. It is also attached to one of the air hoses with a fitted plastic piece attached to the bottom with a “click and lock” like mechanism.

Function

The air component is the main housing unit in which air is compressed. What causes compression is the movement of a piston within the air component which when moved up and down causes changes in air pressure which pushes air into the tire. Also the tire component has a smaller opening at the bottom which concentrates air flow similar to the function of a funnel.

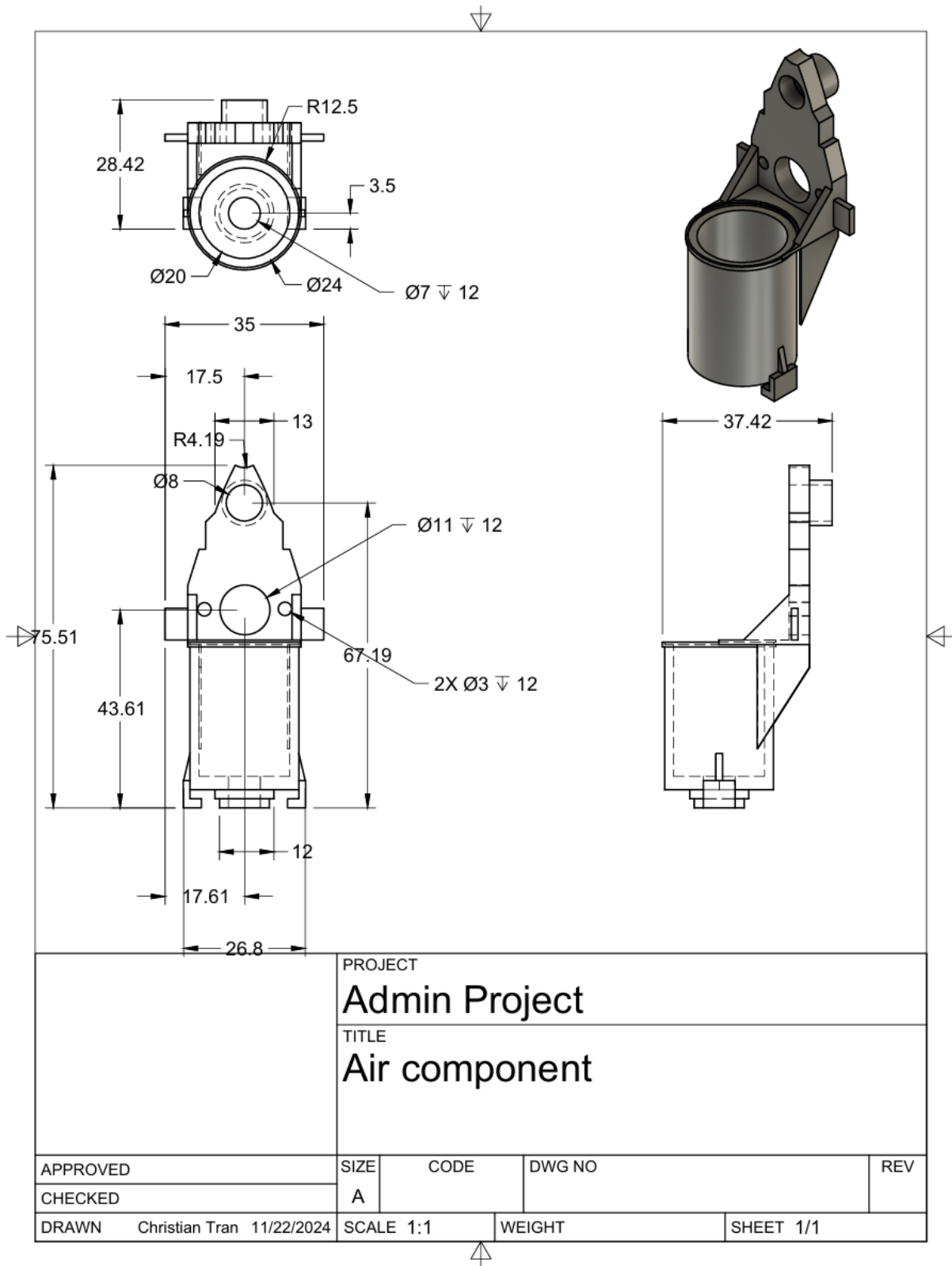


Figure 48 Air Component Dimensioned Drawing

Improvements

Structural/Thermal System:

The first improvement that can be made to the structural/thermal system of the tire pump is to make the gears and pistons out of metal instead of plastic. We observed that the inner components are just resting on holders and are not fastened down. This could cause movement within the system if it is dropped and cause the gear to rub against the outer edge of the plates and cause friction. Even though it is cheaper to make the gears and pistons out of plastic, it is not the best material to use because the longevity of the pump will not hold up as long as the metal will. Switching the gears and pistons to metal will help the pump perform by not wearing down as fast, causing less overheating issues, and taking more time to heat up or potentially melt. By improving these parts of the pump, will pump air more efficiently, will require less maintenance, and will increase the integrity of the pump.

Fluids:

The second improvement that can be made to the tire pump is to increase the volume of the air component and increase the diameter of the piston. By making the tube larger, the pump can hold greater volume, and the increased volume allows the pump to deliver more air per stroke, which will increase the efficiency of the pump. The downside of this is that changes in the gear ratio of the gears or a faster motor will be needed for this to work. However, by changing the sizes of the air components along with changing the gear ratio, the process of compressing a tire would be significantly sped up and more efficient.

Human Interfaces:

The third improvement that can be made to the tire pump is to change the standard analog tire pressure gauge to a digital gauge. By changing the analog gauge to a digital one, it would be

easier for users to read with a simple display, it would be easier to read at night because it is backlit, and it will help people who are unfamiliar with analog pressure gauges. It will also allow the option to display multiple different units. Digital gauges are also highly accurate and provide readings with decimals. Switching the analog tire pressure gauge to a digital one allows for a more efficient and precise tire inflation process that would provide superior performance over the old analog gauge.

Conclusion

After the complete disassembly of the tire pump, we were able to develop a greater understanding of each of the components that take part in the entirety of the tire pump device. After having analyzed each component of the tire pump, it has been made clear of the crucial functions and responsibilities each of these parts has to allow for the tire pump to operate in the way it is designed to. This seemingly simple device has proved to be far more complex than the normal eye would see as it includes various human interface, fluid, structural and thermal components that make up its entirety so that it may simply pump air into flattened automobile tires. Analysis of these components allowed us to determine possible improvements to the various systems. These include making the plastic gear out of metal, increasing the volume of the air component and changing the analog pressure gauge to a digital gauge. The reverse engineering of this electric tire pump has brought light to how in-depth a simple, portable device such as a tire pump can truly be.